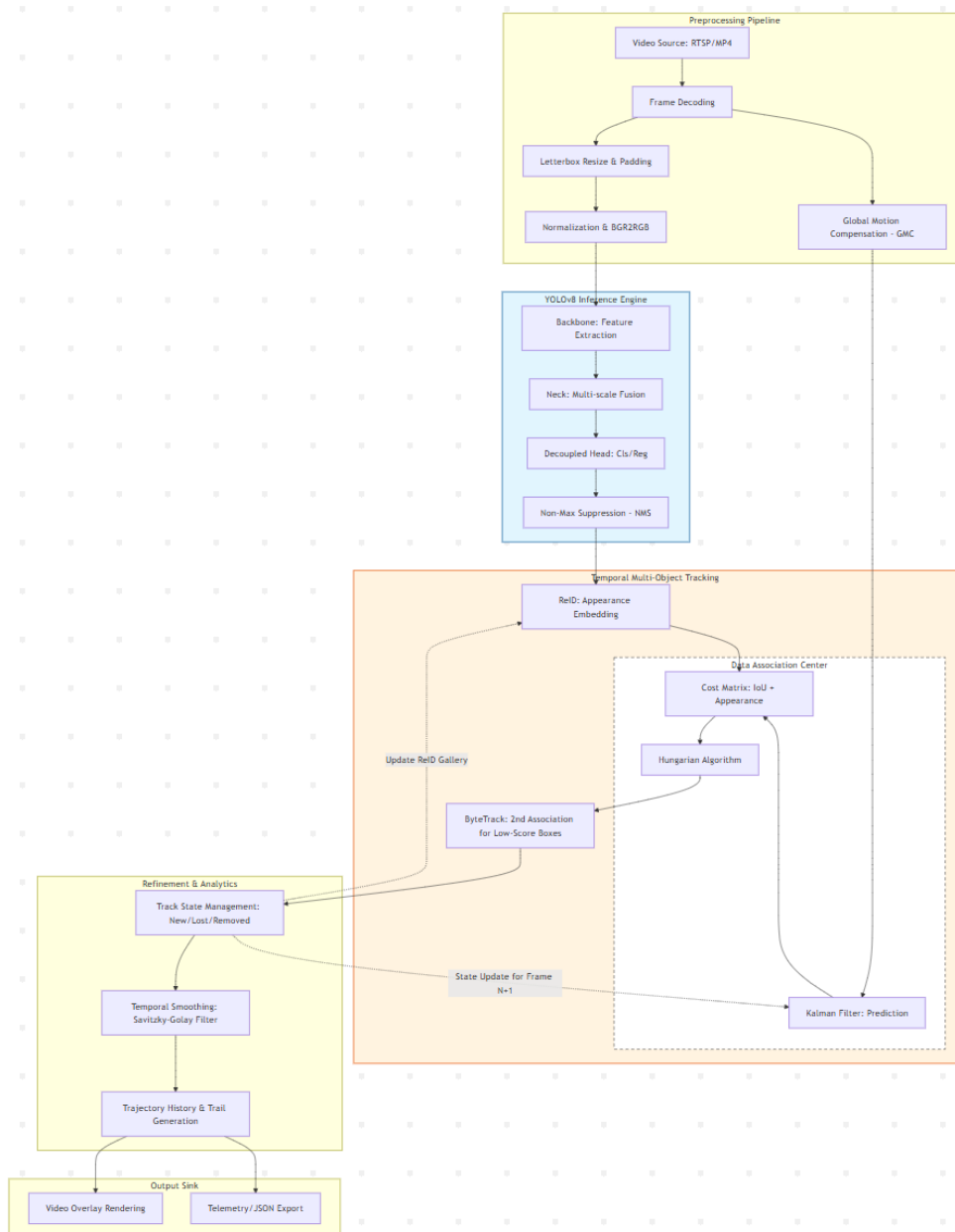


Edge Intelligence Lab

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25MML0025

To design and implement the pipeline for Object Detection in Videos using YOLOv8:



Workflow:

1. **Frame Acquisition & Decoding:** Capture the raw video stream and decode frames into a processable format (BGR/RGB).
2. **GMC Calculation:** Analyze the global motion between the current and previous frame to compensate for camera movement/panning.
3. **Preprocessing:** Apply Letterbox resizing (to maintain aspect ratio), normalization, and color-space conversion for the model.
4. **YOLOv8 Inference:** Run the image through the Backbone, Neck, and Decoupled Head to generate raw bounding boxes.
5. **Post-Inference (NMS):** Apply Non-Maximum Suppression to remove redundant overlapping boxes and keep only high-confidence detections.
6. **Feature Extraction (ReID):** Generate a visual embedding (appearance "fingerprint") for each detected object using a CNN.
7. **Kalman State Prediction:** Predict the "new" location of existing objects from the previous frame based on their last known velocity and GMC data.
8. **First Association (Cost Matrix):** Match high-confidence YOLO detections with Kalman predictions using a combined score of IoU (spatial) and ReID (visual).
9. **Hungarian Matching:** Solve the linear assignment problem to optimally pair detections to existing tracks.
10. **Second Association (ByteTrack):** Attempt to match low-confidence detections (blurred/occluded) with remaining "lost" tracks to prevent ID switches.
11. **Track State Management:** Update track statuses (Mark as New, Tracked, Lost, or Removed if missing for too long).
12. **Temporal Feedback Loop:** Feed the updated track states back into the Kalman Filter and ReID gallery to inform the next frame's prediction.
13. **Smoothing & Refinement:** Apply filters (like Savitzky-Golay) to the box coordinates to eliminate jitter and draw motion trajectories.
14. **Output Sink:** Render the final annotated frames to a video file or stream the metadata (JSON/Telemetry).